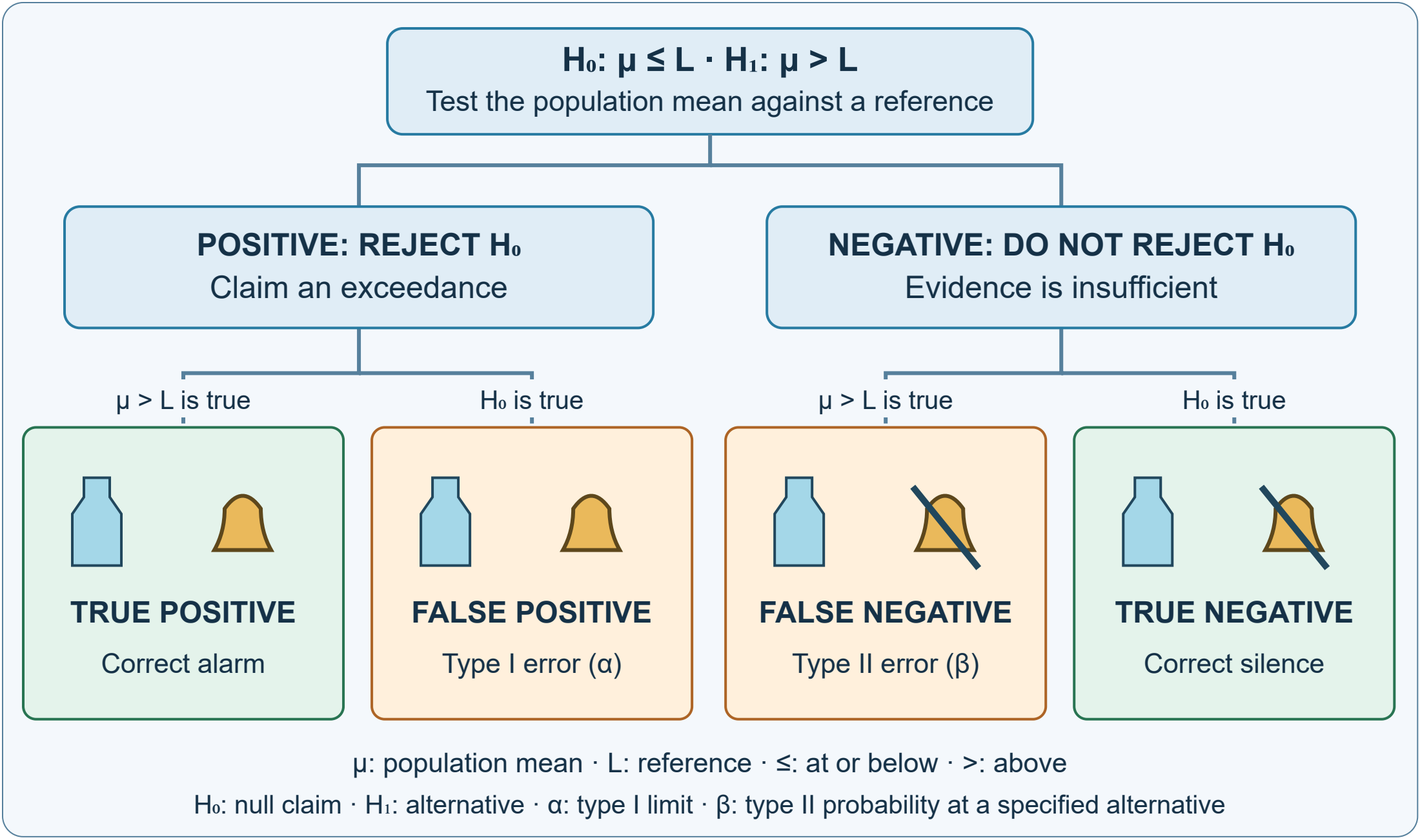

H₀: μ ≤ L · H₁: μ > L
Test the population mean against a reference
POSITIVE: REJECT H₀
Claim an exceedance
NEGATIVE: DO NOT REJECT H₀
Evidence is insufficient
μ > L is true
H₀ is true
μ > L is true
H₀ is true
TRUE POSITIVE
Correct alarm
FALSE POSITIVE
Type I error (α)
FALSE NEGATIVE
Type II error (β)
TRUE NEGATIVE
Correct silence
μ: population mean · L: reference · ≤: at or below · >: above
H₀: null claim · H₁: alternative · α: type I limit · β: type II probability at a specified alternative